

Development of water quality index for groundwater quality evaluation in and around Vemula Mandal, YSR District (A.P., India)

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ABSTRACT

The water quality index method is applied for groundwater quality evaluation in Vemula mandal, YSR District of Andhra Pradesh (South India). For this study, twenty-five groundwater samples were collected in 2018 and analyzed for pH, electrical conductivity, total dissolved solids, total alkalinity, calcium, magnesium, carbonate, bicarbonate, chloride, and fluoride. These parameters were considered to calculate the water quality index (WQI), which ranges between 16.26 and 182.65. Most of this region's groundwater is not appropriate for drinking purposes. Evaluation of groundwater quality from various methods revealed that electrical conductivity, total dissolved solids, total hardness and fluoride, exceed the drinking water quality standards, recommended by World Health Organization. This study focuses on the importance of water quality indexing for groundwater quality assessment which helps to take necessary steps to avoid the use of contaminated water, and take appropriate action for the government and other policy makers in providing safe drinking water to local people in this area.

Key words: Water quality index, Ground water, Vemula mandal, Y.S.R District, A.P., Contaminated water

INTRODUCTION

Groundwater is considered a primary requirement for human beings. It is the most vital, precious resource used for drinking, domestic and irrigation purpose around the world (Sunitha and Sudharshan Reddy, 2019). Because groundwater is of good quality and requires less treatment, it was used more compared to surface water. Groundwater is the only reliable resource due to dwindling surface water bodies (Sunitha et al., 2012 a) and is less susceptible to pollution than surface water and has therefore been used more than fresh water (Laxmi Thockchom and Kshetrimayum, 2019). As the demand for water is increasing day by day due to the rapid growth of industrialization and urbanization, the quality of groundwater is getting significantly depleted. This has become a hectic task in developing countries like India, where people rely on groundwater for drinking purpose. Hence to evaluate the groundwater quantity and quality and providing a database is very important for planning and development of water management techniques.

The Water Quality Index (WQI) method for assessing groundwater quality is an effective tool for the appraisal and management of groundwater and is widely use around the world (Yidana and Yidana , 2010; Ketata-Rokbani et al., 2010; Wu jianhua et al., 2011; AI-hadithi, 2012; Sunitha et al., 2016; Suvarna et al., 2018; Prasad et al., 2019; Narsimha and Sudarshan , 2018). WQI is defined as

the composite influence of different parameters of water quality (Ramkrishnaiah et al., 2009). Horton (1965) established the primary WQI for the first time. Later, Brown et al., (1970) and Deininger and Landwehr (1971) have improved them for river pollution survey carried out in Scotland (Scottish Development Department, 1975). WQI is basically a mathematical equation that transforms large quantities of data on water quality into a single number. In terms of quality and potential uses of any water body, it is the simplest and easiest method for decision-makers. It integrates complex data and generates a score that describes the water quality status. The objective of this work is to evaluate the suitability of groundwater for drinking purposes based on calculated water quality index values.

STUDY AREA

Vemula is situated towards west of Kadapa, Y.S.R District, South India, between 14°21'00'' N and 78° 22'30'' E (Figure 1). It falls into Toposheet No: 57 J/7 and belong to semi-arid region with an average annual rainfall of 600-650 mm. In this region, the temperature ranges from 20.4°C and 43.2°C (C.G.W.B, 2013). Rainfall is the main source of groundwater recharge for drinking and irrigation purposes, and this water is the only source for people in this area.

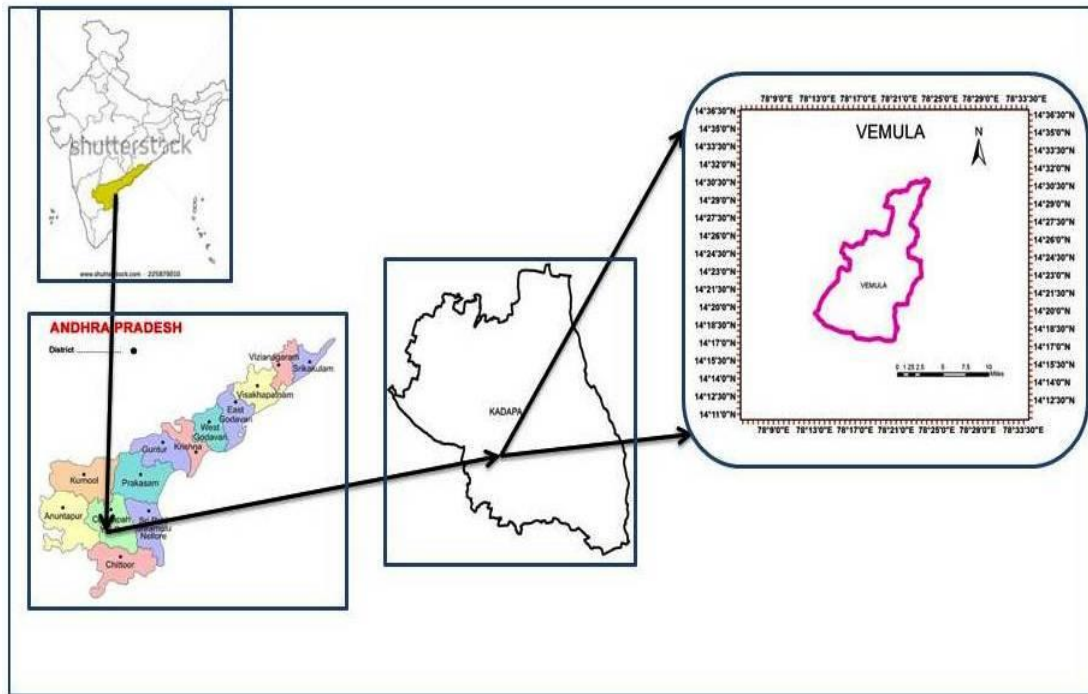


Figure 1. Location map of the Study area

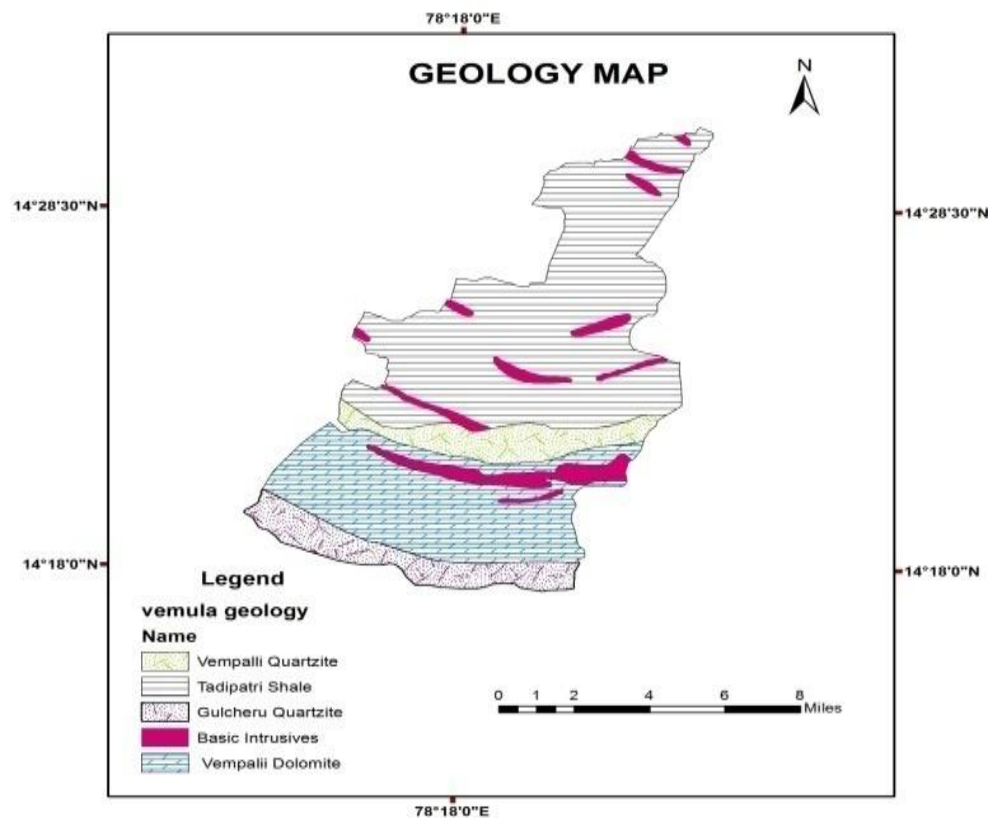


Figure 2. Geological map of the Study area

The major geomorphic units of the study area are Denudational hills, Pediment and Pediplain, structural hills. Paddy, groundnuts, sunflower, sweet lime and vegetables are some of the important crops. Soil types vary from black clays to alluvial, brown and mixed soil types. Geologically, the study area belongs to the lower Cuddapah super group of Papaghni and Chitravathi groups (Nagaraja Rao et al., 1987). The major lithology

observed are conglomerate, shales, basalts, dolomites, limestones, dolomitic limestone, Pupule shale, dolostone (uraniferous) and cherty limestone (Reddy et al., 2012). Geology map of the study area is shown in Figure 2. Groundwater occurs mainly in joints, fractured and fissured zones of quartzites, limestones in both unconfined and semi confined conditions. The water depth varies between 10 and 20 m (C.G.W.B 2007, 2013).

METHODOLOGY

Twenty-five groundwater samples in and around Vemula, were collected in 2018 (Figure 3). Each groundwater sample was analyzed for 11 parameters such as electrical conductivity (EC), pH, total dissolved solids (TDS), calcium, magnesium, bicarbonate, carbonate, chloride and

fluoride adopting standard methods (Hem, 1985; Raghunath, 1987; APHA, 2012). Suggested precautions to avoid contamination have been taken. Determination of pH and EC by pH, conductivity meter, TDS by TDS meter, TH, Ca^{2+} , Mg^{2+} , CO_3^{2-} , HCO_3^- and Cl^- by titrimetry and F^- determination by selective ion electrode (Orion 4 star ion meter, Model: pH/ISE).

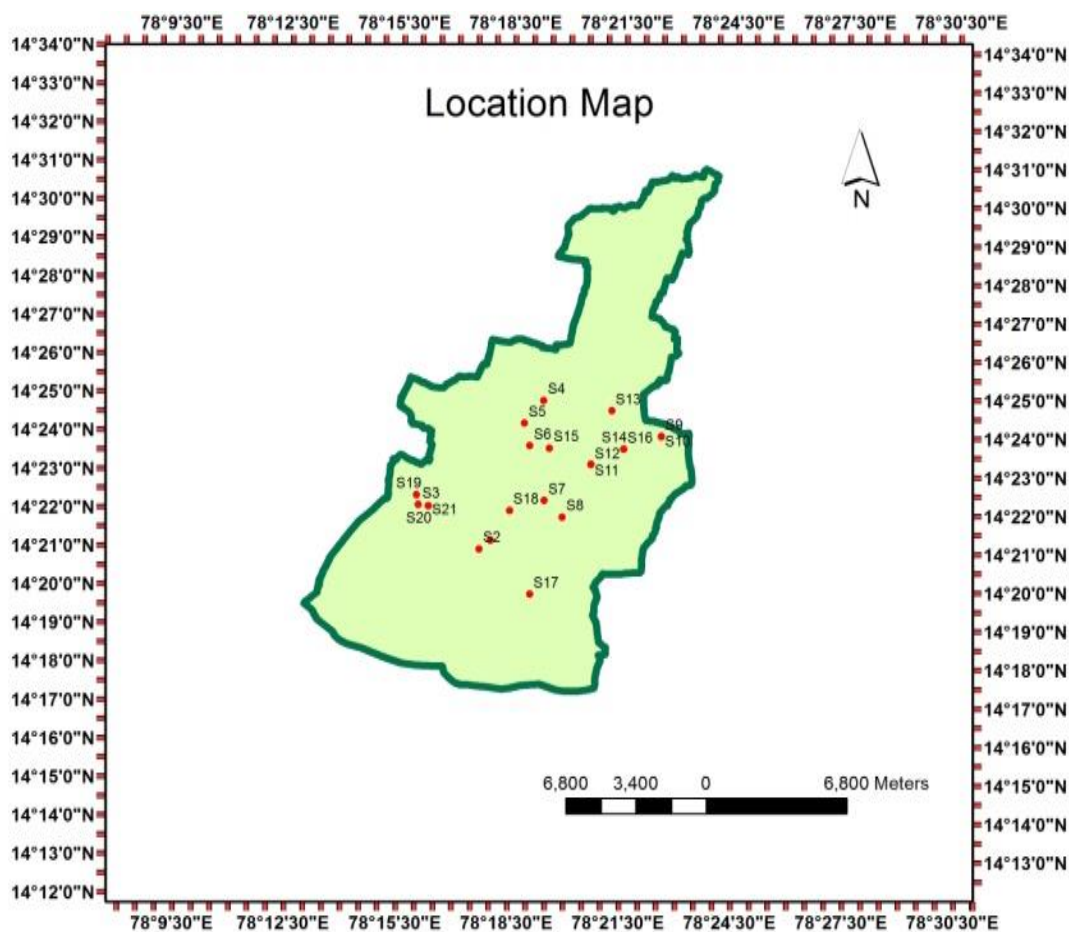


Figure 3. Sample location map of the study area.

RESULTS AND DISCUSSION

The Chemical quality data, as provided in Tables 1 and 2, shows the chemical parameter's statistical summary.

Table 3 provides a comparison between groundwater quality drinking water standards, BIS and WHO.

Table 1. Analytical data for groundwater samples (EC in $\mu\text{S/cm}$, remaining mg /L, except pH)

S.No	Sample ID	pH	EC $\mu\text{S/cm}$	TDS mg/L	TH mg/L	Ca ²⁺ mg/L	Mg ²⁺ mg/L	CO ₃ ²⁻ mg/L	HCO ₃ ⁻ mg/L	TA mg/L	Cl ⁻ mg/L	F ⁻ mg/L
1	V1	8.42	823	382	160	24	34.2	12	292.8	304.8	28.4	0.39
2	V2	8.85	390	203	160	8	9.72	24	73.2	97.2	35.5	0.24
3	V3	8.1	1450	750	300	48	58.32	24	146	170	312.4	0.27
4	V4	8.5	2590	1340	280	16	43.74	24	317.2	341.2	454.4	1.27
5	V5	8.25	369	188	100	40	14.58	12	122	134	42.6	0.626
6	V6	8.8	1500	800	200	16	38.88	36	317.2	353.2	35.5	2.06
7	V7	8.04	2150	1100	280	16	58.32	24	292.8	316.8	78.1	0.73
8	V8	8.4	1040	540	200	16	38.88	12	219.6	231.6	142	0.66
9	V9	8.07	634	327	180	16	19.44	24	195.2	219.2	134.9	0.46
10	V10	7.05	1070	555	220	40	38.88	12	268.4	280.4	21.3	0.79
11	V11	8.39	1430	740	340	24	77.76	36	244	280	220	0.71
12	V12	8.21	1760	910	140	24	0	24	219.6	243.6	134.9	1.6
13	V13	8.67	2170	1120	520	56	116.64	36	244	280	369.2	1.22
14	V14	8.97	1450	750	180	16	9.72	48	366	414	85.2	2.55
15	V15	8.61	1180	610	140	56	24.3	24	317.2	341.2	56.8	3
16	V16	7.87	1710	880	280	16	43.74	60	24.4	84.4	234.3	0.877
17	V17	8.51	826	423	200	40	24.3	24	122	146	71	1.6
18	V18	8.8	1220	630	160	40	14.58	36	268.4	304.4	120.7	0.88
19	V19	8.05	1000	516	260	88	38.88	24	170.8	194.8	106.5	0.8
20	V20	7.95	1150	590	360	40	53.46	12	195.2	207.2	85.2	0.66
21	V21	8.1	811	421	200	40	29.16	24	195.2	219.2	49.7	0.87
22	V22	8.13	1070	550	240	24	43.74	24	47.9	71.9	99.4	0.92
23	V23	7.75	967	500	260	24	38.88	36	38.3	74.3	78.1	0.74
24	V4	8.31	792	409	200	40	24.3	12	38.8	50.8	42.6	0.66
25	V25	8.43	774	400	200	40	29.16	36	40	76	35.5	1.19

Table 2. Summary of physico-chemical parameters in vemula

Parameters	Unit	Minimum	Maximum	Mean	SD	CV	Median
pH		7.05	8.97	8.28	0.40	4.88	8.28
EC	$\mu\text{S/cm}$	369	2590	1211.3	537.3	44.3	1110
TDS	mg/L	188	1340	624.3	279.08	44.6	572.5
TH	mg/L	100	520	233	87.25	37.4	200
Ca ²⁺	mg/L	8	88	32.3	17.5	54.46	28
Mg ²⁺	mg/L	0	116.4	37.57	23.98	63.8	38.8
CO ₃ ⁻	mg/L	12	60	25.8	12.09	46.7	24
HCO ₃ ⁻	mg/L	24.4	366	185.2	106.4	57.4	195.2
Cl ⁻	mg/L	21.3	454.4	121.5	110.5	90.9	85.2
F ⁻	mg/L	0.24	3	1.05	0.67	63.8	0.83

Table 3. Comparison of groundwater quality with standards for drinking water, BIS and WHO

Water quality parameter	W.H.O max accept limit	W.H.O max allow limit	BIS Max accept limit	BIS max allow limit	Concentration in study area	Percent compliance
pH	7.0	8.5	6.5	8.5	7.05-8.97	19%
EC ($\mu\text{S/cm}$)	400	1500	500	1500	369-2590	15%
TDS (mg/L)	500	1500	500	1500	188-1340	Nil
TH (mg/L)	100	500	300	600	100-520	3%
Ca^{2+} (mg/L)	75	200	75	200	8-88	Nil
Mg^{2+} (mg/L)	50	150	30	100	0-116.6	Nil
HCO_3^- (mg/L)	150	300	-	-	24.4-366	15%
Cl^- (mg/L)	200	600	250	1000	21.3-454.4	11%
F^- (mg/L)	0.6	1.5	0.6	1.5	0.24-3	19%

Application of the WQI

In WQI formulation, the importance of different parameters depends on the intended use of water. Water quality parameters are studied from the point of view of suitability for human consumption. The standards used in this study are those recommended by WHO (2011) and the ISI 1983 for the different drinking water parameters. As mentioned before, eleven water quality parameters were considered for calculating the WQI. The water quality Index (WQI) uses the World Health Organization (WHO) and ISI recommended drinking water quality standards.

As shown in the following steps, WQI was calculated using a weighted arithmetic index method (Brown et al., 1970)

Calculation of water quality (q_n):

$$q_n = 100 [(V_n - V_i) / (S_n - V_i)]$$

Where

q_n = n^{th} parameter water quality rating

V_n = The value of the n^{th} parameter observed

S_n = Standard recommended parameter

V_i = ideal value for the n^{th} parameter

All ideal values (V_i) are taken as zero for drinking water except pH 7.0, oxygen dissolved = 14.6 mg / L.

Unit weight calculation (W_n):

Unit weight of the corresponding parameter was calculated by an inverse proportional value to the recommended standard Value S_n .

$$W_n = K / S_n$$

Where

W_n = n^{th} parameter unit weight

S_n = n^{th} parameter standard value

K = constant proportionality

The proportionality constant has been calculated using the equation:

$$K = 1 / \sum (1/S_n)$$

The total water quality index was calculated linearly by aggregating the quality rating with the unit weight

$$WQI = \sum q_n W_n / \sum W_n$$

Table 4. Relative weights for each parameter

Parameter	S_n	$1/S_n$	K	$W_n = K/S_n$	(V_i)	(V_n)	(Q_n)	$W_n Q_n$	WQI
pH	8.5	0.117647	1.240725	0.145968	7	8.42	94.67	13.82	36.68828
EC	1500	0.000667		0.000827	0	823	54.87	0.05	
TDS	1500	0.000667		0.000827	0	382	25.47	0.02	
TH as CaCO_3	500	0.002		0.002481	0	160	32.00	0.08	
Ca^{2+}	200	0.005		0.006204	0	24	12.00	0.07	
Mg^{2+}	150	0.006667		0.008271	0	34.2	22.80	0.19	
Total Alkalinity	200	0.005		0.006204	0	304.8	152.40	0.95	
Cl^-	600	0.001667		0.002068	0	28.4	4.73	0.01	
F^-	1.5	0.666667		0.82715	0	0.39	26.00	21.51	
		0.80598		1				36.69	

Table 5. WQI at every sampling station.

Sample No	WQI	Water quality status
1	36.6888	Excellent
2	19.068	Excellent
3	16.264	Excellent
4	86.491	Good
5	47.401	Excellent
6	132.709	Poor water
7	52.072	Good water
8	51.237	Good water
9	36.8	Excellent
10	45.463	Excellent
11	36.8	Excellent
12	101.426	Poor water
13	57.605	Good
14	161.413	Poor water
15	182.65	Poor water
16	57.357	Good
17	103.826	Poor water
18	63.04	Good
19	55.674	Good
20	67.415	Good
21	97.983	Good
22	62.511	Good
23	48.86	Excellent
24	49.739	Excellent
25	80.232	Good

Table 6. Water quality classification based on WQI value

Class	WQI	Water Quality Status	% of samples in study Area
A	< 50	Excellent	40
B	51-100	Good	40
C	101-200	Poor Water	20
D	201-300	Very Poor Water	Nil
E	> 300	Water Unsuitable For Drinking	Nil

Water quality index of relative weights for each parameter were given in Table 4, WQI at individual sampling station were shown in Table 5. The calculated WQI values range from 16.264 to 182.65 in this study. This can therefore be grouped into five classes: Excellent, Class A (< 50);

Good, Class B (51-100); Poor water, Class C (101-200); Poor water, Class D (201-300); drinking water, Class E (> 300) unsuitable for drinking water. According to the water quality index classification, 40 % of the Vemula region's groundwater samples fall into Class A: Excellent, another

40 % fall in Class B, good and 20 % fall in Class C, poor water category, as shown in Table 6. WQI's high value may be due to higher levels of TDS, EC, TH, and fluoride.

It was found that groundwater pH ranges from 7.0-8.97 (Tables 1 and 2). pH values are within the desirable limits for all samples. At the station in the village of Rangoripalli, a maximum pH of 8.97 was recorded and a minimum pH of 7.05 was recorded at Gondipalli. Most groundwater is observed to be alkaline in nature. Groundwater pH is suitable for drinking and irrigation purposes. The EC ranges from 369 to 2590 $\mu\text{S}/\text{cm}$ at 25°C with an average of 1211 $\mu\text{S}/\text{cm}$ (Table 2).

The highest EC concentration was observed in the southern part of Gollalaguduru village. Some parts of the study area have high EC because of extensive farming practices. Only 3% of the samples exceed the permissible limit of EC (Table 3). In the study area, the TDS value varies from a minimum of 188 mg/L to maximum of 1340 mg/L (Tables 1 and 2), indicating that most groundwater samples are within the

maximum permissible limit. TDS of the groundwater of the area was classified as suggested by the U.S Geological Survey (2000). Table 7 shows that 88.46% of groundwater samples fall within the fresh water category, while 11.5% of groundwater samples fall within the slightly saline category.

High concentration of TDS in the groundwater sample is due to soil salt leaching, and percolation of other mine waste into groundwater, which may lead to increased TDS values. In the present study in the southwestern part of the village of Gollalaguduru, a minimum concentration of TDS 188 mg/L was observed and a maximum concentration of 1340 mg/L was in the village of Gondhipalli, Vempalli. The acceptable limit of total hardness (TH) is 300 mg/L.

Water with TH can be considered as safe <75; moderate to hard: 75-150; hard: 150-300; very hard : >300 (Sawyer et al, 2003), and most groundwater (76.92%) of this study area is considered hard and requires pre-use processing (Table 8).

Table 7. Classification based on TDS: US Geological Survey 2000

Parameter	Classification	Range	Number of samples	% of Samples
TDS	Fresh water	<1000	23	88.46
	Slightly saline	1000-3000	3	11.5
	Moderately saline	3000-10,000	Nil	Nil
	High saline	10,000-35,000	Nil	Nil

Table 8. Groundwater classification based on total hardness

Parameter	Classification	Range	Number of samples	% of Samples	Reference
TH	Safe	<75	Nil	Nil	Sawyer et al. (2003)
	Moderate Hard	75-150	3	11.5	
	Very Hard	150-300	20	76.92	
		>300	3	11.5	

Minimum carbonate concentration 100 mg/L was observed in the present study in the village of Votlapalli and a maximum concentration of 520 mg/L were observed in the Chagaleru, Vemapalli Mandal. Minimum 24 mg/L bicarbonate was observed in the village of Chagaleru and maximum concentration of 366 mg/L was observed in the village Rangoripalli, Vemapalli Mandal. The alkalinity natural water is due to the salts of carbonates,

bicarbonates, borates, silicates and phosphates along with hydroxyl ions in the Free State. In the study area, chloride concentration ranges from 21 mg/L to 454 mg/L with an average of 121 mg/L. In drinking water, the desirable chloride limit is 250 mg/L and the allowable limit is 1000 mg/L (W.H.O, 1990). This can also due to upper soil layer leaching due to industrial domestic activity and dry climate (Narasimha and Sudharshan, 2018).